Important notes Mth-622

Final term for MCQ'S

Made by:- "**VU Life Study**"

1. **Path Independence**

- Independence of path is defined as that the **work done** by an object remains **same** between **initial point** and the **final destination** of the particle, it does not matter which path is taken by the object.
- "A force field is said to **conservative** if the total work done by the particle moving along a curve is **independent** of the path taken by the particle and depend upon the end points of the **curve** only."

1. **Necessary and sufficient conditions for a Conservative Force Field:-**

- A force field F is conservative if and only if there exists a **continuously differentiable** scalar field V such that $\boldsymbol{\vec{F}=-\nabla V}$ or, equivalently, **if and only if** **curl $\vec{F}$= $\nabla\times\vec{F}$=0 identically**.
- A continuously differentiable force field F is **conservative** if and only if for **any closed nonintersecting curve C** (simple closed curve) $W=\oint_c \vec{F}.d\vec{r} = 0$
- The total work done in moving a particle around **any closed path** is **zero.**

2. **Examples of Conservative Forces**

- **Gravitational force** , **Elastic spring force** is example of conservative force.
- The work done of a particle moving along a **closed path is zero** and the force which causes such **motion is conservative**.
- Forces that cannot be expressed in the term of a **potential energy** function are called **non-conservative** forces.
- The forces that **do not store** energy are called **nonconservative** or **dissipative** forces.
- If there is **no scalar** function V such that $F=-\Delta V$ [or, **equivalently, if $\boldsymbol{V\times F=0}$**], then F is called **a non-conservative** force field.
- **Friction,** the impulse (**time dependent** force) is also a non-conservative force.
- **Simple Harmonic Motion (SHM)** is a particular type of **oscillation** and periodic motion in which restoring force of an object is **directly proportional** to the displacement of the object acting in opposite direction of displacement.
- Mathematically, the **restoring force $\boldsymbol{F}$** is given by $F_R = \boldsymbol{-kx}$**.**

  where the **subscript $\boldsymbol{R}$** represents the restoring force and $\boldsymbol{k}$ is the constant of **proportionality** often called the **spring constant** or **modulus of elasticity.**
- In Newtonian mechanics, by Newton's second law we have eq. of S.H.M
- $F = ma = m\dfrac{d^2x}{dt^2} = -kx$ or $\boldsymbol{m\ddot{x}+kx=0}$**.**

- This vibrating system is called a simple harmonic oscillator or linear harmonic oscillator.
- **Mass on a spring:-**
  An object of mass m linked to a spring of spring constant k represents the simple harmonic motion **in closed region.**
- The equation representing the period for the mass attached on a spring
- $\boldsymbol{T} = 2\pi\sqrt{\frac{m}{k}}$
- The above equation expresses that the **time period** of oscillation is **independen**t of **amplitude** as well as the acceleration.
- **Simple pendulum:-**
  The movement of the mass attached to a simple pendulum is considered as simple harmonic motion.
- The time period of a mass m attached to a pendulum of length l with gravitational acceleration g is given by T= $2\pi\sqrt{\frac{l}{g}}$.
- **Amplitude**:-
- Amplitude is defined as **the maximum distance** covered by the oscillating body **in one oscillation or length** of a wave measured from its **mean position**.
- The amplitude of a pendulum is **one-half the distance** that the mass covered in moving from **one terminal to the other**.
- The vibrating sources generate **waves,** whose amplitude is **proportional** to the amplitude of the **vibrating source**.
- Time period is **minimum time** required by a oscillation system to complete **its one cycle** of oscillation of the specific system.
- It is denoted by **T** and measured in **seconds**.
- The **frequency (f)** of an oscillatory system is the number of oscillations pass through a **specific point** in **one second**.
- It is measure in **hertz (Hz).**
- The frequency of **S.H.M** can be calculate by using the following relation $f = \frac{1}{T}$.
- If **T is the kinetic energy**, **V the potential energy** and $\boldsymbol{E = T + V}$ the total energy of a simple harmonic oscillator, then we have $K.E = T = \frac{1}{2}mv^2$

  and $V = \frac{1}{2}kx^2$.

  Then the **total energy of S.H.M** will be $E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$.
- The forces acting on a harmonic oscillator are called **damping forces** which tend to **decrease the amplitude of the successive oscillations** or simply force apposing the motion.
- The damping force is **proportional** to the **velocity.**

- Mathematically, $Fd = -bv = -b\dot{x}$
- where d represents the **damping force** and b is the **damping coefficient**.
- The **negative** sign shows that that direction of Fd is **opposite** to the **velocity v.**
- $\frac{b}{m} = 2\zeta$,
- $\sqrt{\frac{k}{m}} = \omega 0$
- The equation of damped harmonic oscillator can be written as **¨$+2\zeta\dot{x} + \omega_0^2$ x = 0.**
- The theorem, called **Euler's theorem**, is fundamental in the motion of **rigid bodies**.
- "A rotation of a **rigid body** about a **fixed point** of the body is **equivalent** to a rotation about a line **which passes through the point**."
- The line referred to is called **the instantaneous axis of rotation**.
- **Rotations** can be considered as **finite or infinitesimal**.
- **Finite** rotations **cannot** be represented by **vectors** since the **commutative law fails.**
- **Infinitesimal** rotations can be represented by **vectors.**
- **Chasle's theorem** states that the most general **rigid body displacement** can be produced by a **translation along a line** (called its screw axis) followed (or preceded) by a rotation about that line.
- A **rigid** body has **six degrees** of **freedom**.
- By **Euler's theorem**, **three** of these are associated with **pure rotation.**
- The remaining **three** must be associated with **translation**.
- To describe the general motion of a rigid body, think of the general motion as translation of a fixed point O in the body to a point O' followed by the rotation about an axis through O'.
- **Kinematics** is the branch of mechanics deals with the **moving objects** without reference to the forces which cause the **motion**.
- Some **features** of **rigid body motion** are:-
- Displacement.
- Position.
- Velocity.
- Linear Velocity & Angular Velocity.
- Linear Acceleration & Angular Acceleration.
- Motion of a Rigid Body (Translation & Rotation).
- In **Space** this is closely related to the concepts of point, **position**, ' **direction** and **displacement**'. Measurement in space involves the concepts of **length or distance**, with which we assume familiarity.
- **Units** of length are **feet**, **meters**, **miles**, etc.
- In **Time** this concept is derived from our experience of having **one event** taking place **after, before** or **simultaneous with another** event.
- **Units** of time are **seconds, hours, years**, etc.

- In **Matter** Physical objects are composed of **"small bits of matter"** such as **atoms** and **molecules**
- A measure of the **"quantity of matter"** associated with a particle is called its **mass**.
- **Units** of **mass** are **grams, kilograms,** etc.
- Unless otherwise stated we shall assume that the mass of a particle **does not change** with **time.**
- When a **moving particle** remains on a **single straight line**, the motion is said to be **rectilinear.**
- The **general** equation of **motion** is then $= ma \Longrightarrow (x, \dot{x}, \dot{x}) = m\ddot{x}$.
- Rectilinear motion of a body is defined by considering the **two point** of a body covered the **same distance** in the **parallel** direction.
- An **exampl**e of linear motion is an **athlete running** g along a **straight** track.
- The rectilinear motion can be of two types:-
  i. Uniform rectilinear motion.
  ii. Non uniform rectilinear motion.
- **Uniform rectilinear** motion is a type of motion in which the body moves with **uniform velocity** or **zero acceleration.**
- **Non uniform** rectilinear motion is such type of motion with **variable velocity** or **nonzero acceleration.**
- **Uniformly accelerated** rectilinear motion is a special case of **non-uniform** rectilinear motion along a line is that which arises when an object is subjected to **constant** acceleration. This kind of motion is called **uniformly accelerated motion**.
- Uniformly accelerated motion is a type of motion in which the velocity of an object changes by an **equal amount in every equal intervals of time**.
- An example of uniformly accelerated body is **freely falling** object in which the amount of gravitational acceleration **remains same**.
- $F = mg$
- The motion of a particle moving in a **curved path** is called **curvilinear motion**.
- **Example**: A stone thrown into the air at **an angle**.
- **Curvilinear motion** describes the motion of a moving particle that confirms to **a known or fixed curve.** The study of such motion involves the use of **two co-ordinate** systems, the first being **planar motion** and the latter being **cylindrical motion.**
- **Tangential** and **normal** unit vectors are usually denoted by $\vec{e}_t$ *and* $\vec{e}_n$ respectively.
- **Velocity of Curvilinear motion:-**
- If the tangential and normal unit vectors are $\vec{e}_t$ *and* $\vec{e}_n$ respectively, then the velocity will be $\vec{v} = \frac{ds}{dt}\vec{e}_t$

  You have already learnt that $\boldsymbol{v}$ **= vT**.
- **Acceleration of Curvilinear motion**:-
- If the tangential and normal unit vectors are $\vec{e}_t$ *and* $\vec{e}_n$ respectively, then the acceleration will be

- $a^{\rightarrow} = \frac{d^2 s}{dt^2}\vec{e}_t + \frac{(ds/dt)^2}{\rho}\vec{e}_n$
- You have already learnt that
- $a = T\frac{dv}{dt} + \frac{v^2}{r}N$
- Examples of Curvilinear motion:-
- A stone thrown into the air at **an angle.**
- A car driving along a **curved road**.
- **Throwing paper** airplanes or paper **darts** is an example of curvilinear motion.
- $\boldsymbol{e}$**1,2 and** $\boldsymbol{e}$**3** are **mutually perpendicular** and so the cylindrical coordinate system is **orthogonal.**
- **Introduction to Projectile**
- If a ball is thrown from **one person to another** or an object is dropped from a moving plane, then their path of traveling/motion is often called a **projectile.**
- If **air resistance** is **negligible**, a projectile can be considered as a **freely falling body** so the Eq of motion will be $m\frac{d^2 r}{dt^2} = -mg$ or $\frac{d^2 r}{dt^2} = -g$.
- "The law of conservation of energy describes that the **net energy** of an isolated system **remains conserved**. Energy **can neither be created nor destroyed**; rather, it transforms from one form to another."
- In case of conservative force field, the **total energy** is a **constant**.
- If T is for kinetic energy and V is for potential energy, then the total energy E is $E = T + V$ = constant .
- Impulse is a special type of force defined by applying the integral of a force F, over the time interval, t, for which it acts on the body.
- Impulse is a **directional (vector)** quantity in the **same direction** of force as force is also a **directional quantity**.
- When Impulse is applied to a rigid body, it results a **corresponding vector** change in its **linear momentum** along the **same direction**.
- The SI unit of impulse is the **newton second** ($N \cdot s$), and the dimensionally equivalent unit of momentum is the **kilogram meter per second** ($kgms-1$).
- Torque is defined as the turning effect of a body. It is trend of an acting force due to which the rotational motion of a **body changes**. It is also called **twist** and **rotational force** on an object.
- Torque is defined as the cross product of the force vector to the distance vector, which causes rotational motion of the body. $\boldsymbol{\tau} = \vec{r} \times \vec{F}$
- The magnitude of **torque** depends upon the **applied force.**
- The length of the lever arm connecting the axis to the point where the force applied, and the angle between the force vector and the length of lever arm. Symbolically we can write it as: $\boldsymbol{\tau} = |\boldsymbol{r}||\boldsymbol{F}|\sin\boldsymbol{\theta}$

- Torque is a **vector quantity** implies that it has direction as well as **magnitude**.
- The SI unit for torque is the **newton meter (Nm).**
- The **direction** of torque can be approximate using **Right Hand Rule.**

- ## Rigid bodies:-
- When a force is applied to an object/ system of particles, and if the object maintains its **overall shape**, then the object is called a **rigid body.**
- Gap between **two fixed points** on the rigid body remains **same regardless** of external forces exerted on it.
- We can neglect the **deformation** of such bodies.
- A rigid body usually has **continuous** distribution of mass.
- ## Definition of Elastic Bodies:-
- When a force is applied to a system of particles, **it changes the distance** between individual particles. Such systems are often called **deformable** or elastic bodies.
- Examples
- A **spring** and **rubber band** are some common examples of **elastic bodies.**
- A **wheel** is a common example of **rigid body**.
- **Properties of Rigid Bodies**
- The number of coordinates required to specify the position of a system of **one or more particles** is called the number of **degrees of freedom** of the system. For example a particle moving freely in space requires 3 coordinates, e.g. (x, y, z), to specify its position. Thus the number of **degrees of freedom is 3.**
- A displacement of a **rigid body** is a direct change of position of its particles.
- Translational motion is the displacement of all particles of the body by the same amount and the line segment joining **the initial and the final position** of the particles represented by parallel vectors. Examples of translational motion are particles freely falling down to earth and the motion of a bullet fired from a gun.
- Circular motion of a body about a **fixed point** or axis is called **rotation.**
- If during a displacement the points of the rigid body on some line **remains fixed** and all other are **displaced** through the **same angle**, then this displacement is called **rotation**.
- A rigid performs rotations around **an imaginary line** called a rotation axis.
- If the axis of rotation passes through the **center of mass** of the rigid body then body is said to be **spin** or **rotate upon itself.**
- If a body rotates about **some external fixed point** is called **revolution** or **orbital motion** of the rigid body.
- The example of revolution is the rotation of **earth around sun and motion of moon around sun.**
- Rotational motion **concerns** only with **rigid bodies**. The reverse rotation of a body (inverse rotation) is also a rotation.
- A **wheel** is common examples of rotation.

- Translational motion along the given **fixed plane** and **rotational motion** about a **suitable** axis **perpendicular** to the plane.
- This fixed axis is specifically chosen to pass through the **center of mass** of the rigid body.
- The axis about which the rigid body rotates is called **instantaneous axis of rotation**, where this axis is **perpendicular** to the plane.
- The point **where instantaneous axis meets the fixed plane** along which the body performs translation motion is described as the instantaneous **center of rotation.**
- The center of mass (c.m.) or centroid of system of particles is a **hypothetical particle** such that if the entire mass of the system were concentrated there, the **mechanical** properties would **remain the same.**
- In particular expression of linear momentum, angular momentum and kinetic energy assume **simpler or more convenient forms** when referred to the coordinated of this hypothetical particle and the equation of motion can be reduced to simpler equation of a **single particle.**
- If a system **experiences no external force**, the center-of-mass of the system will remain **at rest**, or will move at **constant velocity** if it is already moving.
- If there is an **external force**, the center of mass accelerates according to $\boldsymbol{F = ma}$**.**
- Basically, the center-of-mass of a system can be treated as a **point mass**, following Newton's Laws.
- If an object is **thrown into the air**, different parts of the object can follow **quite complicated paths**, but the center-of-mass will follow a **parabola.**
- If an object explodes, the different pieces of the object will follow seemingly **independent paths** after the explosion. The center of mass, however, will keep doing what it was doing before the explosion. This is because an explosion involves only **internal forces**.
- The moment of inertia of a rigid body is a property which depends upon its **mass and shape**, (i.e. the mass distribution of the body) and determines its behavior in rotational motion.
- In rotational motion, the moment of inertia plays the **same** role as the mass in linear motion.
- Formally the moment of inertia I of the particle of mass m about a line is defined by $I = md^2$ where d is the perpendicular distance between the particle and the line (called the axis).
- Radius of gyration specifies the distribution of the elements of body around the axis in terms of the mass moment of inertia.
- When a rigid body is rotating about a fixed point O, the **angular velocity** vector ω and the **angular momentum** vector L (about O) **are not in general in the same** direction.
- Each point in the body there exists distinct directions, which are fixed relative to the body, along which the **two vectors** are **aligned** i.e. **coincident.** Such directions are called **principal directions** and the axes along them are referred to as **principal axes of inertia.**
- The corresponding moments of inertia are called principal moments of inertia.

- If the principal axes at each point of the body exist, then their **orthogonality** can be proved by stating that axes relative to which product of inertia are **zero** are the principal axes.
- Dynamics is the branch of mechanics deals **with forces and relationship** fundamentally to the motion but sometimes also to the **equilibrium of bodies.**
- These will be two types of forces, **internal and external**. Internal forces act between particles of the system; all other are **external**.
- The rigid bodies are a system of particles in which the position of particles is **relatively fixed.**
- An actual rigid body consists of **very large number** of particles and therefore we may suppose that there is a **continuous distribution** of mass.
- If (r) denotes density of a volume element dV, surrounding the point r, then the mass of this element will be $(r)V$.
- The **angular momentum** of a single particle is defined as the **cross product** of linear momentum and position vector of concerned particle. Mathematically $= r \times mv$.
- The time rate change of angular momentum in the absence of some external forces is **zero**. Mathematically, we can write $\boldsymbol{dL/dt = 0 \Longrightarrow L =}$ **constant**.
- The time **rate of change of the angular** momentum of a system is equal to the total moment of all the **external forces** acting on the system.
- If a system is isolated, then $\boldsymbol{N = 0}$, and the **angular momentum** remains **constant** in both **magnitude** and **direction**:
- Kinetic energy is the energy produced in any body during **its motion**. It is equal to the **half** of the product of **mass and square** of the velocity of the moving body.
- **Solid Circular Cylinder** :-
- We assume the radius of cylinder is $\boldsymbol{a}$ and mass M about axis of cylinder $I = \frac{1}{2}Ma^2$.
- **Hollow Circular Cylinder:-**
- We assume the radius of cylinder is $\boldsymbol{a}$ and mass M about axis of cylinder. We consider the thickness of Wall of cylinder is negligible. $I = Ma^2$
- **Solid Sphere:-**
- We assume the radius of sphere is $\boldsymbol{a}$ and mass M about a diameter. $I = \frac{2}{5}Ma^2$.
- **Hollow Sphere:-**
- We assume the radius of sphere is $\boldsymbol{a}$ and mass M about a diameter. We consider the thickness of sphere is negligible. $I = Ma^2$
- **Rectangular Plate:-**
- We consider sides of length $\boldsymbol{a}$ and $\boldsymbol{b}$, and mass M about an axis perpendicular to the plate through the center of mass. $I = \frac{1}{12}M(a^2 + b^2)$
- **Thin Rod:-**

- We assume the length of rod is $\boldsymbol{a}$ and mass M about an axis perpendicular to the rod through the center of mass. $I = \frac{1}{12} Ma^2$
- **Triangular Lamina:-**
- We assume the height $\boldsymbol{h}$ and mass M of lamina. $I = \frac{1}{6} Mh^2$.
- **Right Circular Cone:-**
- We assume the radius of the circular cone is a and mass M. $I = \frac{3}{10} Ma^2$.
- The moment of inertia of a plane rigid body about an axis perpendicular to the body is equal to the sum of the moment of inertia about two mutually perpendicular axis lying in the plane of the body and meeting at the common point with the given axis.
- The axes relative to which **product** of inertia are **zero** are called the principal axes and the moment of inertia along these axes are called **principal moment of inertia**.
- For a rigid body, there exist a set of **three mutually orthogonal** axes called principal axes relative to which the product of inertia are **zero** $\vec{\omega}$ and L are considered along the **same** direction.
- This system will have a **non-trivial** solution, i.e. ($\vec{\omega} \neq 0$), if the determinant of the matrix of coefficients is **zero.**
- A 3×3 symmetrical matrix has **three** real eigenvalues, which may be **distinct** or **repeated.**
- The eigenvectors of a **symmetrical** matrix corresponding to distinct eigenvalues are **orthogonal.**
- It is always possible to find three mutually **orthogonal** eigenvectors for a 3×3 symmetric matrix, whether the eigenvalues are **distinct or repeated.**
- The principal of moment of inertia are **always real** numbers. This is obviously physical because the moment of inertia is defined as the quantity $\sum_i m_i d_i^2$ where mi and di are both **real.**
- When all **three** principal moments I_1, I_2, I_3 are **distinct**, then by the theorem 2, three mutually **orthogonal principal** axes can be found.
- When $I_1 = I_2$ but $I_1 \neq I_3$, then by the theorem 3, we can still determine **three mutually orthogonal** principal axes, because the inertia matrix is **symmetric**.
- In many instances a body possesses **sufficient symmetry** so that at least **one principal** axis can be found by inspection, i.e;the axis can be chosen so as to make **two of the three** products of inertia **vanish.**
- We consider a plane **rigid body** i.e. a 2D body; for example a plate of **uniform thickness**. Such a system can be regarded as a **coplanar distribution** of mass.
- Since there are **three mutually orthogonal** principal axes, one of them must be **perpendicular** to the plane of the body.
- The other **two axes** will lie in the plane of **lamina**.

- Suppose a rigid body has **no axis** of symmetry. Even so, the **tensor** that represents the moment of inertia of such a body, is characterized by a **real, symmetric 3×3** matrix that can be **diagonalized.** The resulting diagonal elements are the values of the principal moments of inertia of the rigid body.

- The axes of the coordinate system, in which this matrix is diagonal, are the principal axes of the body, because all products of inertia have **vanished.**
- Thus, finding the principal axes and corresponding moments of inertia of any rigid body, **symmetric or not**, is virtually the same as to **diagonalzing** its moment of inertia matrix.
- The solutions are **not independent**. They are subject to the constraint $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$
- In other words the resultant vector e_1 specified by these components is a **unit vector**.

Thank You..!
Remember me in ur prayers.